SLIDE 1–2 | Antibacterial Drugs (Lectures 3, 4, 5)

- **Course map: A.** Beta-Lactam & Other Cell Wall- & Membrane-Active Antibiotics
- **B.** Protein Synthesis Inhibitors (Aminoglycosides & Spectinomycin, Tetracyclines, Macrolides, Clindamycin, Chloramphenicol, Streptogramins, Oxazolidinones)
- **C.** Anti Folate Drugs (Sulfonamides, Trimethoprim) **D.** DNA Gyrase Inhibitors (Quinolones)
- *This PDF = Part A only: cell wall synthesis inhibitors.*

SLIDE 3 | Mechanism of Antibiotic Action

- **Five targets** in a bacterium:
  - Protein synthesis inhibitors → **ribosome**
  - Nucleic acid synthesis inhibitors → **DNA**
  - **Cell wall synthesis inhibitors** → cell wall (**circled = this topic**)
  - Cell membrane integrity disruptors → cell membrane
  - Metabolic pathway inhibitors → **folic acid**

SLIDE 5 | Learning Objectives

- 1. Classify the β-Lactam antibiotics and names of drugs
- 2. Penicillins and other β-lactam antibiotics overview
- 3. Penicillin β-lactam classification and names of drugs
- 4. Pharmacodynamics and mechanism of action
- 5. Pharmacokinetics (ADME)
- 6. Antibacterial spectrum of activity and clinical applications
- 7. Mechanisms of resistance
- 8. Adverse effects

SLIDE 6 | Beta-Lactam & Other Cell Wall- & Cell Membrane-Active Antibiotics

- Some antimicrobial drugs **selectively** interfere with synthesis of the bacterial cell wall — a structure that **mammalian cells do not possess**.
- Cell wall is composed of a polymer called **peptidoglycan**: **glycan units joined to each other by peptide cross-links**.
- To be maximally effective, inhibitors of cell wall synthesis require **actively proliferating** microorganisms; little or no effect on bacteria that are not growing and dividing.
- Most important members: **β-lactam antibiotics** (named after the **β-lactam ring**, essential to activity) and **vancomycin**.

EXPLAIN / REMEMBER

Why selective? Human cells have no cell wall, so these drugs hit bacteria but spare us = **selective toxicity**. **Why need growing bacteria?** The wall is built only during growth/division, so a resting bug is not hit.

SLIDE 7 | Cell envelope of a Gram-negative bacterium

- **Outer membrane** (lipid bilayer): present in **Gram-negative, NOT Gram-positive**.
- Penetrated by **porins** = proteins forming channels that give **hydrophilic access** to cytoplasmic membrane.
- **Peptidoglycan layer** is unique to bacteria; **much thicker in Gram-positive** than Gram-negative.
- Outer membrane + peptidoglycan layer together = the **cell wall**.
- **Penicillin-binding proteins (PBPs)** = membrane proteins that **cross-link peptidoglycan**.
- **β-lactamases**, if present, sit in the **periplasmic space** or on outer surface of cytoplasmic membrane and may destroy β-lactams that get through the outer membrane.

SLIDE 8 | Cell Membrane / Peptidoglycan structure

- Cell wall functions: **1.** Maintain shape [provide rigidity] **2.** Protect bacteria from **osmotic pressure changes**.
- **Peptidoglycans** = glycan + peptide crosslinks. **Glycans: NAG** (N-acetylglucosamine) and **NAM** (N-acetylmuramic acid).
- **Gram +** : **50–100** glycan layers. **Gram –** : **1–2** glycan layers, with **porin** and **periplasmic space**.

EXPLAIN / REMEMBER

Easy way: Gram-positive = thick brick wall (50–100 layers) so no outer membrane needed; Gram-negative = thin wall (1–2 layers) *plus* outer membrane with porins.

SLIDE 9 | Cell Wall Synthesis — THREE STAGES

- **1. Cytoplasmic stage:** synthesis of precursors (NAG, NAM) → inhibited by **cycloserine and fosfomycin**.
- **2. Membrane stage (Elongation and transfer):** transfer of precursors from cytosol to membrane and incorporation into growing peptidoglycan → **bacitracin, vancomycin**.
- **3. Extracellular stage (Cross-linking):** cross-linking of linear peptidoglycan chains by membrane-bound **transpeptidases** → **Beta-lactams**.

EXPLAIN / REMEMBER

Memorize as a table: Stage 1 (cytoplasm) = Cycloserine + Fosfomycin | Stage 2 (membrane) = Bacitracin + Vancomycin | Stage 3 (outside) = β-lactams. Order = inside → membrane → outside.

SLIDE 10 | Stage 1: Synthesis of precursors

- **Glucose → NAG**; UTP converts to **UDP-NAG**.
- UDP-NAG + **phosphoenolpyruvate** via **enol pyruvate transferase** (**inhibited by FOSFOMYCIN**) → **UDP-NAM**.
- UDP-NAM + **L-Ala, D-Glu, L-Lys → UDP-NAM-tripeptide**.
- **L-Ala → (Racemase; inhibited by CYCLOSERINE) → D-Ala → D-Ala-D-Ala** dipeptide.
- Tripeptide + D-Ala-D-Ala → **UDP-NAM-pentapeptide (UDP-NAMpp)**.

EXPLAIN / REMEMBER

Racemase = enzyme that flips L-alanine into D-alanine. Bacteria need D-Ala-D-Ala for the cross-link later, and cycloserine blocks this (it is an analogue of D-Ala). Fosfomycin blocks the step making **NAM**.

SLIDE 11–12 | Stage 2: Elongation and Transfer

- UDP-NAMpp + UDP-NAG carried across membrane by lipid carrier **BACTOPRENOL**.
- **Bacitracin** inhibits the **phosphatase** (removes a phosphate group from a protein): **Dephosphorylation recycles Bactoprenol**.
- **Peptidoglycan synthase** adds the units to the growing chain; **Vancomycin prevents this transfer**.
- **Peptidoglycan** = large polymer, mesh-like scaffold around the bacterial cytoplasmic membrane. Synthesis is vital at several stages of cell cycle: **expansion of scaffold during cell elongation** and **formation of a septum during cell division**.

EXPLAIN / REMEMBER

Bactoprenol = a taxi that carries building blocks across the membrane. After dropping them, it must lose its phosphate (dephosphorylation) to be reused. **Bacitracin** blocks this recycling; **vancomycin** blocks the polymerization (transglycosylation).

SLIDE 13 | Stage 3: Cross-linking

- Neighboring peptidoglycan chains are cross-linked through their **peptide side chains**.
- Reaction is catalyzed by **transpeptidase**. **Transpeptidase = a penicillin binding protein (PBP)**.
- **Beta-lactams inhibit transpeptidase.**

SLIDE 14 | Cross-link: E. coli vs S. aureus

- **Transpeptidase catalyzes a peptide–peptide bond** between the **penultimate D-Ala** and **L-Gly (or Lys)**. The **terminal D-Ala is lost** in the process.
- **E. coli:** L-Lys joins directly to D-Ala of the next chain.
- **S. aureus:** L-Lys → **Gly-Gly-Gly-Gly-Gly** (pentaglycine bridge) → D-Ala.

SLIDE 15–16 | Gram-negative vs Gram-positive; PBP + β-lactam ring

- Diagram: peptidoglycan = **NAG – NAM – oligopeptide**; PBPs act on it. Gram-negative has **outer membrane, lipoproteins, lipopolysaccharides, porin, periplasmic space**; Gram-positive has thick peptidoglycan.
- **β-lactam ring forms a permanent covalent bond with the active site of PBP** (transpeptidase enzyme) → cross-linking blocked.
- **β-Lactam antibiotics = Penicillins, Cephalosporins, Carbapenems, Monobactams.**

EXPLAIN / REMEMBER

Covalent = permanent (irreversible): the drug acts like a key that breaks off in the lock (the enzyme can never work again). This is why the drug is bactericidal.

SLIDE 17 | β-Lactamase and its inhibitors

- **β-lactamase** attacks (hydrolyzes) the β-lactam ring of the antibiotic.
- **β-Lactamase inhibitors: Avibactam, Clavulanic acid, Sulbactam, Tazobactam.**
- Carbapenems and monobactams (β-lactam ring structures) shown as other β-lactams.

SLIDE 18 | CELL WALL INHIBITORS & OTHERS (Classification)

- **Cell wall inhibitors (β-lactam antibiotics):** Penicillins; Cephalosporins; Carbapenems (Imipenem/Cilastatin); Monobactams (Aztreonam).
- **β-lactamase inhibitors:** Clavulanic acid; Sulbactam; Tazobactam.
- **Others:** Vancomycin; Bacitracin.